

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1** The photograph shows a saffron crocus plant.



(Source: © NatureOnline / Alamy Stock Photo)

- (a) This plant grows from a bulb.

The bulb contains starch as an energy storage molecule for the crocus.

How many of the following statements are correct for starch in living cells?

- starch consists of a mixture of two types of polysaccharide
- starch contains only 1–6 bonds
- starch is insoluble in water

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three



- (b) Saffron is a spice that is derived from parts of the flower of the saffron crocus.

Saffron contains a disaccharide called gentiobiose.

- (i) Read through the following description of some disaccharides.

Complete the description by writing the most appropriate word on the dotted lines.

(4)

Disaccharides consist of two monosaccharides joined together by a

..... covalent bond.

Sucrose and lactose are both disaccharides. They both contain a molecule

of Sucrose also contains one molecule of

..... and lactose also contains one molecule of

.....

- (ii) Gentiobiose is formed from two identical monosaccharides.

Name the type of reaction that joins these monosaccharides together in the formation of gentiobiose.

(1)

.....



(iii) The molecular mass of each of the monosaccharides in gentiobiose is 180.

The table shows the molecular mass of the elements present in the monosaccharides.

Element	Molecular mass
carbon	12
hydrogen	1
oxygen	16

Which is the molecular mass of gentiobiose?

(1)

- ☐ **A** 144
- ☐ **B** 162
- ☐ **C** 342
- ☐ **D** 360

(Total for Question 1 = 7 marks)

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